Flutter on OpenHarmony: A New Embedder Direction

Davide Ricci, Vice President, Linaro's Solutions and Services

HDC 2026 - Shenzhen

Abstract

Flutter's architecture wasn't designed for OpenHarmony's ecosystem (Cangjie rendering, ArkUI framework, HAP packaging) making it fundamentally incompatible with the existing Android embedder. Based on a joint Huawei-Linaro engineering review, we evaluated three directions and converged on a pragmatic path: leveraging the Flutter Web engine compiled to Cangjie, bridged to device capabilities via native APIs. This approach respects OpenHarmony's architecture while maximizing code reuse. We'll cover the evaluation, architecture, Phase 1 results, and the upcoming roadmap — including a Phase 2 kick-off in Q3 2026. We'll close by reflecting on Linaro's unique position as a bridge between open-source communities, the Flutter ecosystem, and industry players — supporting Huawei's own divisions (Consumer, Compute, Cloud, Networking) and other OpenHarmony-adopting OEMs with adaptation, development, testing, compliance (CRA, ISO 27001), deployment and long-term support.